

MATH 551: NUMERICAL ORDINARY DIFFERENTIAL EQUATIONS

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1. AUGUST 24, 2026

1.1. **Ordinary differential equations.** An ODE has the form

$$x'(t) = f(t, x(t)).$$

Given an initial condition $x(t_0) = x_0$, the corresponding *initial-value problem* (IVP) is to find $x(t)$ for $t \geq t_0$ such that

$$(1) \quad x'(t) = f(t, x(t)), \quad x(t_0) = x_0.$$

Topics arising in the numerical solution of differential equations include

- non-stiff problems and stiff solvers;

- geometric solvers;
- parameter estimation and Bayesian approaches, where the vector field may depend on a model parameter θ , as in $f(t, x, \theta)$;
- boundary-value problems; and
- stochastic differential equations (SDEs).

We generally consider vector-valued solutions and vector fields,

$$x: \mathbb{R} \longrightarrow \mathbb{R}^n, \quad f: \mathbb{R} \times \mathbb{R}^n \longrightarrow \mathbb{R}^n,$$

which can be written componentwise as

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}, \quad f(t, x) = \begin{bmatrix} f_1(t, x_1, \dots, x_n) \\ \vdots \\ f_n(t, x_1, \dots, x_n) \end{bmatrix}.$$

1.2. Examples.

Example 1. *The scalar equation*

$$x' = x$$

is an ODE with $n = 1$.

Example 2 (Damped oscillator). *Consider the second-order equation*

$$x'' = -\omega^2 x - x'.$$

Setting $x_1 = x$ and $x_2 = x'$ converts it into the first-order system

$$\begin{cases} x_1' = x_2, \\ x_2' = -\omega^2 x_1 - x_2. \end{cases}$$

Thus, with $\mathbf{x} = (x_1, x_2)^\top$, we have $\mathbf{x}' = \mathbf{f}(x_1, x_2)$.

Example 3 (Chemical kinetics). *A three-species kinetic model is*

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad \begin{cases} x_1' = -k_1 x_1 + k_2 x_2 x_3, \\ x_2' = k_1 x_1 - k_2 x_2 x_3 - k_3 x_2^2, \\ x_3' = k_3 x_2^2, \end{cases}$$

where $k_1, k_2, k_3 > 0$ are reaction rates. The reaction rates may occur on very different scales, which can lead to stiffness.

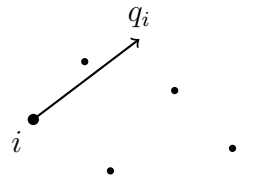
Example 4 (Molecular model). *Let q_i be the position of particle i , let m_i be its mass, and let $V(q_1, \dots, q_N)$ be the potential energy. Newton's equation is*

$$m_i q_i'' = - \frac{\partial V(q_1, \dots, q_N)}{\partial q_i}.$$

Introducing the momentum $p_i = m_i q_i'$ gives

$$\begin{cases} q_i' = \frac{p_i}{m_i}, \\ p_i' = - \frac{\partial V}{\partial q_i}. \end{cases}$$

The positions and momenta can therefore be collected into the state vector

$$\mathbf{x} = \begin{bmatrix} q_1 \\ p_1 \\ q_2 \\ p_2 \\ \vdots \\ q_N \\ p_N \end{bmatrix}.$$


1.3. Solutions and the Lipschitz condition. Consider the IVP (1) on an interval containing t_0 .

Definition 1. *A function $x(t)$ is a solution of the IVP if it satisfies the integral equation*

$$x(t) = x_0 + \int_{t_0}^t f(s, x(s)) ds.$$

Definition 2 (Lipschitz condition). *The vector field f is Lipschitz continuous in x on a domain D if there is a constant $L \geq 0$ such that*

$$\|f(t, x) - f(t, y)\| \leq L\|x - y\|$$

for all $x, y \in D$ and all relevant t .

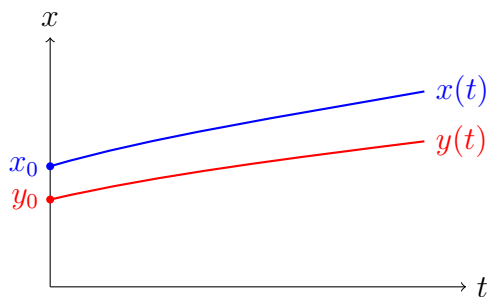
Under the usual continuity assumption in t , the Lipschitz condition in x implies that the solution of the IVP exists and is unique on an appropriate interval.

The same condition also controls the sensitivity to the initial data. Let $x(t)$ and $y(t)$ be solutions with initial values x_0 and y_0 , respectively. Their integral equations give

$$\begin{aligned} \|x(t) - y(t)\| &\leq \|x_0 - y_0\| + \int_{t_0}^t \|f(s, x(s)) - f(s, y(s))\| ds \\ &\leq \|x_0 - y_0\| + L \int_{t_0}^t \|x(s) - y(s)\| ds. \end{aligned}$$

By Grönwall's inequality,

$$(2) \quad \|x(t) - y(t)\| \leq \|x_0 - y_0\| e^{L(t-t_0)}.$$



Thus nearby initial values produce solutions whose separation is bounded by (2).

2. AUGUST 26, 2026

The main topics are properties of solutions, time discretization, and consistency error. We continue to consider the IVP

$$(3) \quad x'(t) = f(t, x(t)), \quad x(t_0) = x_0,$$

where $x: \mathbb{R} \rightarrow \mathbb{R}^n$.

2.1. Stability of solutions. The integral formulation of (3) is

$$x(t) = x_0 + \int_{t_0}^t f(s, x(s)) ds.$$

If $y(t)$ is another solution with initial value y_0 , then

$$y(t) = y_0 + \int_{t_0}^t f(s, y(s)) ds,$$

and hence

$$y(t) - x(t) = y_0 - x_0 + \int_{t_0}^t (f(s, y(s)) - f(s, x(s))) ds.$$

Suppose f is Lipschitz continuous in x with Lipschitz constant L , so that

$$\|f(t, y) - f(t, x)\| \leq L\|y - x\|.$$

Taking norms gives

$$(4) \quad \|y(t) - x(t)\| \leq \|y_0 - x_0\| + \int_{t_0}^t L\|y(s) - x(s)\| ds.$$

The following inequality controls this accumulation of error.

Lemma 1 (Grönwall's inequality). *Let $u, v \in C([t_0, t_1], \mathbb{R})$ be nonnegative functions and let $\rho \geq 0$. If*

$$(5) \quad u(t) \leq \rho + \int_{t_0}^t v(s)u(s) ds, \quad t \in [t_0, t_1],$$

then

$$(6) \quad u(t) \leq \rho \exp \left(\int_{t_0}^t v(s) ds \right), \quad t \in [t_0, t_1].$$

In particular, $u \equiv 0$ when $\rho = 0$.

Proof. First suppose that $\rho > 0$ and define

$$U(t) = \rho + \int_{t_0}^t v(s)u(s) ds.$$

Then $U \in C^1([t_0, t_1], \mathbb{R})$, and

$$u(t) \leq U(t), \quad U(t) \geq \rho > 0.$$

Since u and v are nonnegative,

$$U'(t) = v(t)u(t) \leq v(t)U(t).$$

Thus the logarithmic derivative of U satisfies

$$(\log U(t))' = \frac{U'(t)}{U(t)} \leq v(t).$$

Integrating from t_0 to t yields

$$\log U(t) - \log U(t_0) \leq \int_{t_0}^t v(s) ds.$$

Because $U(t_0) = \rho$ and $u(t) \leq U(t)$, applying the exponential function proves (6) when $\rho > 0$.

Now suppose that $\rho = 0$. For any $\varepsilon > 0$, (5) implies

$$u(t) \leq \varepsilon + \int_{t_0}^t v(s)u(s) ds.$$

The first part of the proof, with $\rho = \varepsilon$, gives

$$0 \leq u(t) \leq \varepsilon \exp \left(\int_{t_0}^t v(s) ds \right).$$

Letting $\varepsilon \downarrow 0$ shows that $u(t) = 0$ for every $t \in [t_0, t_1]$. □

Apply Lemma 1 to (4) with

$$u(t) = \|y(t) - x(t)\|, \quad v(t) = L, \quad \rho = \|y_0 - x_0\|.$$

We obtain

$$(7) \quad \|y(t) - x(t)\| \leq \|y_0 - x_0\| e^{L(t-t_0)}.$$

This estimate can be pessimistic, but it establishes continuous dependence on the initial data.

2.2. The solution map. The *solution map*, or *flow map*, is denoted by

$$x(t) = \Phi^{t,t_0}(x_0).$$

For example, consider the scalar logistic equation

$$x' = x(1 - x).$$

For $x_0 \neq 0$, its solution is

$$\Phi^{t,t_0}(x_0) = \frac{1}{1 + \frac{1-x_0}{x_0}e^{-(t-t_0)}}.$$

When $x_0 = 0$, the solution is $x(t) \equiv 0$.

The solution map has the identity and composition properties

$$\begin{aligned}\Phi^{t_0,t_0}(x_0) &= x_0, & \Phi^{t_0,t_0} &= I, \\ \Phi^{t_2,t_1}(\Phi^{t_1,t_0}(x_0)) &= \Phi^{t_2,t_0}(x_0).\end{aligned}$$

The second property is the basis of time marching.

Differentiating the solution map with respect to time gives

$$\begin{aligned}\frac{d}{dt}\Phi^{t,t_0}(x_0) &= f(t, x(t)), \\ \frac{d^2}{dt^2}\Phi^{t,t_0}(x_0) &= f_t(t, x(t)) + f_x(t, x(t))x'(t) \\ &= (f_t + f_x f)(t, x(t)),\end{aligned}$$

where f_x is the Jacobian matrix of f with respect to x . Higher time derivatives can be found by repeated differentiation, for example with automatic differentiation; they can also be viewed through the Koopman operator.

To measure sensitivity with respect to the initial value, define the Wronskian matrix

$$W(t) = \frac{\partial}{\partial x_0}\Phi^{t,t_0}(x_0) = \frac{\partial x(t)}{\partial x_0}.$$

Differentiating with respect to t gives

$$\begin{aligned}W'(t) &= \frac{\partial}{\partial x_0}f(t, x(t)) \\ &= f_x(t, x(t))\frac{\partial x(t)}{\partial x_0}.\end{aligned}$$

Thus, if

$$A(t) = f_x(t, x(t)),$$

then W satisfies the *variational equation*

$$(8) \quad W'(t) = A(t)W(t), \quad W(t_0) = I.$$

2.3. One-step methods. Choose a time grid

$$t_0 < t_1 < \cdots < t_N = t, \quad \tau_j = t_{j+1} - t_j,$$

where the step sizes τ_j are small. Repeated use of the composition property gives

$$\Phi^{t_N, t_0} = \Phi^{t_N, t_{N-1}} \circ \cdots \circ \Phi^{t_2, t_1} \circ \Phi^{t_1, t_0}.$$

It is therefore enough to approximate a single short-time solution map. Writing $t_j = t$ and $t_{j+1} = t + \tau$, Taylor expansion gives

$$(9) \quad \Phi^{t+\tau, t}(x) = x + \tau f(t, x) + \frac{\tau^2}{2} (f_t + f_x f)(t, x) + O(\tau^3).$$

Dropping the terms of order τ^2 and higher defines the Euler one-step map

$$\Psi^{t+\tau, t}(x) = x + \tau f(t, x).$$

Consequently, Euler's method is

$$(10) \quad x_{j+1} = x_j + \tau_j f(t_j, x_j).$$

Its one-step consistency error follows from (9):

$$\Phi^{t+\tau, t}(x) - \Psi^{t+\tau, t}(x) = \frac{\tau^2}{2} (f_t + f_x f)(t, x) + O(\tau^3) = O(\tau^2).$$

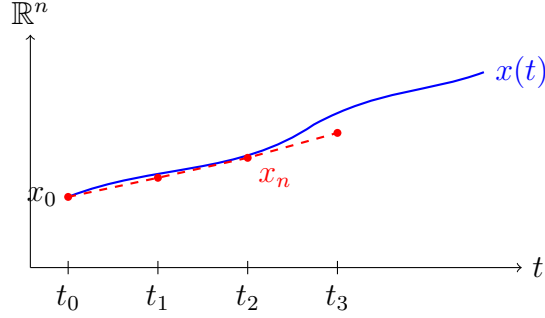
3. AUGUST 28, 2026

We continue the study of one-step methods, focusing on consistency, examples, and convergence. Let

$$t_0 < t_1 < \cdots < t_N, \quad \tau_n = t_{n+1} - t_n,$$

where t_n is a time level and τ_n is the corresponding step size. We use $x(t_n)$ for the exact solution and x_n for its numerical approximation:

$$x(t_n) \approx x_n.$$



A one-step method advances the numerical solution by a map

$$(11) \quad x_{n+1} = \Psi^{t_{n+1}, t_n}(x_n).$$

Thus the exact and numerical solutions at t_n can be written as

$$\begin{aligned} x(t_n) &= \Phi^{t_n, t_{n-1}} \circ \dots \circ \Phi^{t_2, t_1} \circ \Phi^{t_1, t_0}(x_0), \\ x_n &= \Psi^{t_n, t_{n-1}} \circ \dots \circ \Psi^{t_2, t_1} \circ \Psi^{t_1, t_0}(x_0). \end{aligned}$$

3.1. Methods obtained from quadrature. The integral form of the ODE over one step is

$$(12) \quad x(t_{n+1}) = x(t_n) + \int_{t_n}^{t_{n+1}} f(s, x(s)) ds.$$

Different quadrature approximations of this integral produce different one-step methods.

3.1.1. Forward Euler method. Using the left endpoint in (12) gives

$$x(t_{n+1}) \approx x(t_n) + \tau_n f(t_n, x(t_n)).$$

Replacing the exact solution values by numerical approximations yields

$$(13) \quad x_{n+1} = x_n + \tau_n f(t_n, x_n) = \Psi_{\text{FE}}^{t_{n+1}, t_n}(x_n).$$

3.1.2. Backward Euler method. Using the right endpoint instead gives

$$x(t_{n+1}) \approx x(t_n) + \tau_n f(t_{n+1}, x(t_{n+1})).$$

The resulting method is

$$(14) \quad x_{n+1} = x_n + \tau_n f(t_{n+1}, x_{n+1}) = \Psi_{\text{BE}}^{t_{n+1}, t_n}(x_n).$$

Because x_{n+1} occurs on both sides, backward Euler requires the solution of a fixed-point problem at each step.

3.1.3. *A predictor–corrector method.* The trapezoidal rule suggests

$$x(t_{n+1}) \approx x(t_n) + \frac{\tau_n}{2} (f(t_n, x(t_n)) + f(t_{n+1}, x(t_{n+1}))).$$

We can first predict the right endpoint with forward Euler and then use that prediction in the trapezoidal formula:

$$(15) \quad x_{n+1}^* = x_n + \tau_n f(t_n, x_n),$$

$$(16) \quad x_{n+1} = x_n + \frac{\tau_n}{2} (f(t_n, x_n) + f(t_{n+1}, x_{n+1}^*)).$$

This explicit two-stage scheme is also called Heun’s method. Together, the two stages define a one-step map

$$x_{n+1} = \Psi_H^{t_{n+1}, t_n}(x_n).$$

3.2. **Local consistency error.** For a numerical one-step map Ψ , define its *local error*, or *consistency error*, at (t_n, x) by

$$(17) \quad \mathcal{E}_n(x) = \Phi^{t_{n+1}, t_n}(x) - \Psi^{t_{n+1}, t_n}(x).$$

Here the exact and numerical steps start from the same value x at time t_n .

Example 5 (Forward Euler). *The Taylor expansion from (9) gives*

$$\begin{aligned} \mathcal{E}_n^{\text{FE}}(x) &= \Phi^{t_n + \tau_n, t_n}(x) - (x + \tau_n f(t_n, x)) \\ &= \frac{\tau_n^2}{2} (f_t + f_x f)(t_n, x) + O(\tau_n^3) \\ &= O(\tau_n^2). \end{aligned}$$

Example 6 (Backward Euler). *Let*

$$y(\tau) = \Psi_{\text{BE}}^{t_n + \tau, t_n}(x).$$

The defining fixed-point equation is

$$y(\tau) = x + \tau f(t_n + \tau, y(\tau)).$$

Differentiating this equation and evaluating at $\tau = 0$ gives

$$y'(0) = f(t_n, x), \quad y''(0) = 2(f_t + f_x f)(t_n, x).$$

Therefore,

$$\Psi_{\text{BE}}^{t_n + \tau_n, t_n}(x) = x + \tau_n f(t_n, x) + \tau_n^2 (f_t + f_x f)(t_n, x) + O(\tau_n^3).$$

Comparing this with the exact flow expansion yields

$$\begin{aligned} \mathcal{E}_n^{\text{BE}}(x) &= -\frac{\tau_n^2}{2} (f_t + f_x f)(t_n, x) + O(\tau_n^3) \\ &= O(\tau_n^2). \end{aligned}$$

3.3. Global error and convergence. The *global error* at t_n is

$$(18) \quad e_n = x_n - x(t_n).$$

The local errors accumulate as the method marches forward in time. On a fixed time interval there are approximately $1/\tau$ steps of size τ , so one generally loses one power of τ when passing from local to global error:

$$\mathcal{E}_n = O(\tau^{p+1}) \implies e_n = O(\tau^p).$$

More generally, with

$$h = \max_{0 \leq n < N} \tau_n,$$

a stable one-step method with local error $O(h^{p+1})$ has global error $O(h^p)$ under the usual smoothness assumptions. Thus both forward and backward Euler have local error of order two and global order one.

4. AUGUST 31, 2026

We continue the analysis of one-step methods. The main goals are to pass from local consistency to global convergence and to introduce Runge–Kutta methods. Consider the IVP

$$x'(t) = f(t, x(t)), \quad x(t_0) = x_0 \in \mathbb{R}^n,$$

on a grid

$$t_0 < t_1 < \cdots < t_N = T, \quad \tau_j = t_{j+1} - t_j, \quad h = \max_{0 \leq j < N} \tau_j.$$

4.1. From local consistency to global convergence. Recall that Φ^{t_{j+1}, t_j} denotes the exact flow over one step, while Ψ^{t_{j+1}, t_j} denotes the numerical one-step map. For a starting value x , define the local consistency error by

$$(19) \quad \xi_j(x) = \Phi^{t_{j+1}, t_j}(x) - \Psi^{t_{j+1}, t_j}(x).$$

Definition 3 (Consistency order). *A one-step method has consistency order $p > 0$ if, uniformly for the relevant starting values,*

$$\|\xi_j(x)\| \leq C\tau_j^{p+1}$$

as $h \rightarrow 0$. The order p need not be an integer when the data have only Hölder regularity.

Forward and backward Euler have consistency order one. The exact and numerical values at the final time can be written as compositions:

$$\begin{aligned} x(T) &= \Phi^{t_N, t_{N-1}} \circ \cdots \circ \Phi^{t_2, t_1} \circ \Phi^{t_1, t_0}(x_0), \\ x_N &= \Psi^{t_N, t_{N-1}} \circ \cdots \circ \Psi^{t_2, t_1} \circ \Psi^{t_1, t_0}(x_0). \end{aligned}$$

Thus the local errors are propagated through all the subsequent steps.

Let

$$e_j = x_j - x(t_j)$$

be the global error. For brevity, set $\Psi_j = \Psi^{t_{j+1}, t_j}$ and $\Phi_j = \Phi^{t_{j+1}, t_j}$. Then

$$\begin{aligned} e_{j+1} &= \Psi_j(x_j) - \Phi_j(x(t_j)) \\ (20) \quad &= \Psi_j(x_j) - \Psi_j(x(t_j)) - \xi_j(x(t_j)). \end{aligned}$$

Suppose the one-step map satisfies the stability estimate

$$(21) \quad \|\Psi_j(y) - \Psi_j(z)\| \leq e^{L\tau_j} \|y - z\|.$$

Combining the consistency and stability estimates in (20) gives

$$(22) \quad \|e_{j+1}\| \leq e^{L\tau_j} \|e_j\| + C\tau_j^{p+1}.$$

Since $e_0 = 0$, iteration yields

$$\begin{aligned} \|e_N\| &\leq C \sum_{j=0}^{N-1} \tau_j^{p+1} e^{L(T-t_{j+1})} \\ &\leq Ch^p \sum_{j=0}^{N-1} \tau_j e^{L(T-t_{j+1})}. \end{aligned}$$

For $L > 0$, the sum is bounded by a geometric integral, and hence

$$(23) \quad \|e_N\| \leq \frac{C}{L} (e^{L(T-t_0)} - 1) h^p.$$

When $L = 0$, the corresponding bound is

$$\|e_N\| \leq C(T - t_0) h^p.$$

In either case, $h \rightarrow 0$ implies $e_N \rightarrow 0$. Thus stability converts a local error of order $p + 1$ into a global error of order p .

Remark 1 (Contractive problems). In the favorable case

$$\|\Psi_j(y) - \Psi_j(z)\| \leq \|y - z\|,$$

no exponential amplification occurs. Direct summation gives

$$\|e_N\| \leq C \sum_{j=0}^{N-1} \tau_j^{p+1} \leq C(T - t_0) h^p.$$

Such contractivity is especially relevant for dissipative ODEs and numerical methods that preserve their dissipative structure.

4.2. Runge–Kutta methods. Consider an explicit method with s stages and coefficients a_{ij} , b_i , and c_i . Starting from x at time t with step size τ , its stages are computed successively from

$$(24) \quad k_i = f \left(t + c_i \tau, x + \tau \sum_{j=1}^{i-1} a_{ij} k_j \right), \quad i = 1, \dots, s,$$

and its one-step map is

$$(25) \quad \Psi^{t+\tau, t}(x) = x + \tau \sum_{i=1}^s b_i k_i.$$

Example 7 (Forward Euler). Take $s = 1$, $c_1 = 0$, $a_{11} = 0$, and $b_1 = 1$. Then

$$k_1 = f(t, x), \quad \Psi^{t+\tau, t}(x) = x + \tau k_1,$$

which is forward Euler, a first-order method.

Example 8 (Explicit trapezoidal method). Take $s = 2$ and

$$A = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \quad b = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}, \quad c = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

The stages and update are

$$\begin{aligned} k_1 &= f(t, x), \\ k_2 &= f(t + \tau, x + \tau k_1), \\ \Psi^{t+\tau, t}(x) &= x + \frac{\tau}{2}(k_1 + k_2). \end{aligned}$$

This is the predictor–corrector method from the previous lecture, also called Heun’s method. It has order two.

5. SEPTEMBER 2, 2026

We develop the coefficient representation of Runge–Kutta methods and derive their order conditions. Throughout, we retain the notation of the previous lectures: $\tau_n = t_{n+1} - t_n$, x_n is the numerical approximation at t_n , Φ is the exact flow, and Ψ is the numerical one-step map.

5.1. Butcher tableaux. An s -stage Runge–Kutta method has the form

$$(26) \quad k_i = f \left(t_n + c_i \tau_n, x_n + \tau_n \sum_{j=1}^s a_{ij} k_j \right), \quad i = 1, \dots, s,$$

$$(27) \quad x_{n+1} = x_n + \tau_n \sum_{i=1}^s b_i k_i.$$

The coefficients are collected into

$$A = (a_{ij})_{i,j=1}^s, \quad b = \begin{bmatrix} b_1 & \cdots & b_s \end{bmatrix}^\top, \quad c = \begin{bmatrix} c_1 & \cdots & c_s \end{bmatrix}^\top.$$

They are conventionally displayed in a *Butcher tableau*:

$$\begin{array}{c|cccc}
 c_1 & a_{11} & a_{12} & \cdots & a_{1s} \\
 c_2 & a_{21} & a_{22} & \cdots & a_{2s} \\
 \vdots & \vdots & \vdots & \ddots & \vdots \\
 c_s & a_{s1} & a_{s2} & \cdots & a_{ss} \\
 \hline
 & b_1 & b_2 & \cdots & b_s
 \end{array}$$

The values $t_n + c_i\tau_n$ are the intermediate stage times between t_n and t_{n+1} . If A is strictly lower triangular, then each stage depends only on previously computed stages. The resulting method is an *explicit Runge–Kutta method*.

5.2. The internal consistency relation. Any nonautonomous ODE can be converted into an autonomous system by augmenting its state. Set

$$z(t) = \begin{bmatrix} t \\ x(t) \end{bmatrix}, \quad z'(t) = \begin{bmatrix} 1 \\ f(t, x(t)) \end{bmatrix} = F(z(t)).$$

Apply a Runge–Kutta method to this autonomous system. Because the time component of every stage slope is one, the time component of stage i is

$$t_n + \tau_n \sum_{j=1}^s a_{ij}.$$

For this to agree with the prescribed stage time $t_n + c_i\tau_n$ in (26), we require

$$c_i = \sum_{j=1}^s a_{ij}, \quad i = 1, \dots, s.$$

Thus the *internal consistency relation* is

$$(28) \quad c = A\mathbf{1}, \quad \mathbf{1} = \begin{bmatrix} 1 & \cdots & 1 \end{bmatrix}^T.$$

Under this relation, applying the Runge–Kutta scheme directly to the nonautonomous equation is equivalent to applying it to the augmented autonomous system.

5.3. One-step error and order. For an exact starting value x at time t , define the one-step defect by

$$(29) \quad \mathcal{E}(x, t, \tau) = \Phi^{t+\tau, t}(x) - \Psi^{t+\tau, t}(x).$$

A Runge–Kutta method has order p if

$$\mathcal{E}(x, t, \tau) = O(\tau^{p+1}) \quad \text{as } \tau \rightarrow 0.$$

Assuming (28), it suffices to derive the conditions for an autonomous equation

$$x' = f(x).$$

We write its exact and Runge–Kutta flows as Φ^τ and Ψ^τ , respectively.

Let $f' = Df(x)$ and $f'' = D^2f(x)$. Differentiation along an exact solution gives

$$\begin{aligned} x' &= f, \\ x'' &= f'f, \\ x''' &= f''(f, f) + f'f'f. \end{aligned}$$

Consequently,

$$(30) \quad \Phi^\tau(x) = x + \tau f + \frac{\tau^2}{2} f'f + \frac{\tau^3}{6} (f''(f, f) + f'f'f) + O(\tau^4).$$

For the numerical method, set

$$\Delta_i = \tau \sum_{j=1}^s a_{ij} k_j, \quad k_i = f(x + \Delta_i).$$

Since $k_j = f + O(\tau)$ and $c = A\mathbf{1}$,

$$\Delta_i = \tau c_i f + O(\tau^2).$$

A more detailed Taylor expansion gives

$$k_i = f + \tau c_i f'f + \tau^2 \left(\sum_{j=1}^s a_{ij} c_j f'f'f + \frac{c_i^2}{2} f''(f, f) \right) + O(\tau^3).$$

Substitution into $\Psi^\tau(x) = x + \tau \sum_i b_i k_i$ yields

$$\Psi^\tau(x) = x + \tau(b^\top \mathbf{1})f + \tau^2(b^\top c)f'f$$

$$(31) \quad + \tau^3 \left((b^\top A c) f' f' f + \frac{1}{2} (b^\top c^{\circ 2}) f''(f, f) \right) + O(\tau^4),$$

where $c^{\circ 2} = (c_1^2, \dots, c_s^2)^\top$.

Comparing (30) and (31) gives the order conditions through order three:

$$(32) \quad b^\top \mathbf{1} = 1, \quad (\text{order one}),$$

$$(33) \quad b^\top c = \frac{1}{2}, \quad (\text{additional condition for order two}),$$

$$(34) \quad b^\top c^{\circ 2} = \frac{1}{3}, \quad (\text{additional condition for order three}),$$

$$(35) \quad b^\top A c = \frac{1}{6}, \quad (\text{additional condition for order three}).$$

Therefore (32) gives a first-order method; the first two conditions give a second-order method; and all four give a third-order method. A fourth-order method must satisfy eight conditions in total. For nonautonomous equations, the relation $c = A\mathbf{1}$ is also needed for these autonomous order conditions to apply directly.

5.4. Examples.

Example 9 (Forward Euler). *Forward Euler has the tableau*

$$\begin{array}{c|c} 0 & 0 \\ \hline & 1 \end{array}.$$

Thus $s = 1$, $A = 0$, $b = 1$, and $c = 0$. It satisfies $b^\top \mathbf{1} = 1$, but not the second-order condition, so it is first order.

Example 10 (Explicit trapezoidal method). *The explicit trapezoidal method, or Heun's method, is*

$$\begin{aligned} k_1 &= f(t_n, x_n), \\ k_2 &= f(t_n + \tau_n, x_n + \tau_n k_1), \\ x_{n+1} &= x_n + \frac{\tau_n}{2} (k_1 + k_2). \end{aligned}$$

Its tableau is

$$\begin{array}{c|cc} 0 & 0 & 0 \\ 1 & 1 & 0 \\ \hline & \frac{1}{2} & \frac{1}{2} \end{array}.$$

Here $c = A\mathbf{1}$, $b^\top \mathbf{1} = 1$, and $b^\top c = 1/2$. However, $b^\top c^{\circ 2} = 1/2 \neq 1/3$, so the method is second order but not third order.

5.5. Explicit three-stage methods. A general explicit three-stage Runge–Kutta method has tableau

$$\begin{array}{c|ccc} 0 & 0 & 0 & 0 \\ c_2 & a_{21} & 0 & 0 \\ c_3 & a_{31} & a_{32} & 0 \\ \hline & b_1 & b_2 & b_3 \end{array}.$$

Internal consistency requires

$$c_2 = a_{21}, \quad c_3 = a_{31} + a_{32}.$$

Before imposing the order conditions, the six free coefficients are

$$b_1, b_2, b_3, a_{21}, a_{31}, a_{32}.$$

The four conditions (32)–(35) enforce third order, so one generically expects a two-parameter family of explicit three-stage schemes of order three.

6. SEPTEMBER 4, 2026

We continue the study of Runge–Kutta methods, first constructing families of explicit methods and then introducing embedded methods and stability functions. We use the notation established in the previous lectures: $\tau_n = t_{n+1} - t_n$, x_n is the numerical solution, and Φ and Ψ denote the exact and numerical flows.

6.1. Runge–Kutta methods and order conditions. Recall that an s -stage Runge–Kutta method is

$$k_i = f \left(t_n + c_i \tau_n, x_n + \tau_n \sum_{j=1}^s a_{ij} k_j \right), \quad i = 1, \dots, s,$$

$$x_{n+1} = x_n + \tau_n \sum_{i=1}^s b_i k_i.$$

Here s is the number of stages, $A = (a_{ij})$ contains the stage coefficients, c_i is the i -th stage node, and b_i is the i -th quadrature weight. The internal consistency relation is

$$c = A\mathbf{1}.$$

It makes the method agree on the original nonautonomous equation and its autonomous augmentation. The method is explicit precisely when A is strictly lower triangular.

For the autonomous equation $x' = f(x)$, a method has order p when

$$\|\Phi^\tau(x) - \Psi^\tau(x)\| = O(\tau^{p+1}) \quad \text{as } \tau \rightarrow 0.$$

The first four order conditions, derived in the previous lecture, are

$$\begin{aligned} b^\top \mathbf{1} &= 1, & b^\top c &= \frac{1}{2}, \\ b^\top c^{\circ 2} &= \frac{1}{3}, & b^\top A c &= \frac{1}{6}. \end{aligned}$$

They are sufficient through order three. To obtain order four, four more conditions are needed:

$$\begin{aligned} b^\top c^{\circ 3} &= \frac{1}{4}, & b^\top (c \circ (Ac)) &= \frac{1}{8}, \\ b^\top A(c^{\circ 2}) &= \frac{1}{12}, & b^\top A^2 c &= \frac{1}{24}, \end{aligned}$$

where $\circ$ denotes componentwise multiplication. Thus one, two, four, and eight conditions must be satisfied for orders one, two, three, and four, respectively.

Forward Euler uses one stage and has order one, while the explicit trapezoidal method uses two stages and has order two. We next continue the construction of higher-order explicit methods.

6.2. Explicit three-stage methods of order three. For an explicit three-stage method, internal consistency gives the tableau

$$\begin{array}{c|ccc} 0 & 0 & 0 & 0 \\ c_2 & a_{21} & 0 & 0 \\ c_3 & a_{31} & a_{32} & 0 \\ \hline & b_1 & b_2 & b_3 \end{array}, \quad c_2 = a_{21}, \quad c_3 = a_{31} + a_{32}.$$

There are six free coefficients and four order conditions, so the third-order methods generically form a two-parameter family. Let

$$c_2 = \alpha, \quad c_3 = \beta.$$

Solving the four order conditions gives

$$\begin{array}{c|ccc} 0 & 0 & 0 & 0 \\ \alpha & \alpha & 0 & 0 \\ \beta & \frac{\beta(3\alpha(1-\alpha)-\beta)}{\alpha(2-3\alpha)} & \frac{\beta(\beta-\alpha)}{\alpha(2-3\alpha)} & 0 \\ \hline & \frac{6\alpha\beta-3\alpha-3\beta+2}{6\alpha\beta} & \frac{3\beta-2}{6\alpha(\beta-\alpha)} & \frac{2-3\alpha}{6\beta(\beta-\alpha)} \end{array},$$

provided the displayed denominators are nonzero.

A useful subfamily has $\beta = 1$. Its weights are

$$b_1 = \frac{3\alpha-1}{6\alpha}, \quad b_2 = \frac{1}{6\alpha(1-\alpha)}, \quad b_3 = \frac{2-3\alpha}{6(1-\alpha)}.$$

All three weights are nonnegative when $\alpha \in [1/3, 2/3]$, but positivity of the weights alone does not guarantee that one method is better than another.

Example 11 (Kutta's third-order method). *Taking $\alpha = 1/2$ and $\beta = 1$ gives*

$$\begin{array}{c|ccc} 0 & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 1 & -1 & 2 & 0 \\ \hline & \frac{1}{6} & \frac{2}{3} & \frac{1}{6} \end{array}.$$

Thus

$$k_1 = f(t_n, x_n),$$

$$\begin{aligned}
k_2 &= f\left(t_n + \frac{\tau_n}{2}, x_n + \frac{\tau_n}{2}k_1\right), \\
k_3 &= f\left(t_n + \tau_n, x_n + \tau_n(-k_1 + 2k_2)\right), \\
x_{n+1} &= x_n + \frac{\tau_n}{6}(k_1 + 4k_2 + k_3).
\end{aligned}$$

6.3. Explicit four-stage methods of order four. Before imposing internal consistency, a general explicit four-stage tableau contains thirteen coefficients: three nontrivial nodes, six strictly lower-triangular entries of A , and four weights. Internal consistency removes three degrees of freedom, and the eight fourth-order conditions generically leave two free parameters.

In the usual nonconfluent family one may take

$$c_1 = 0, \quad c_2 = \alpha, \quad c_3 = \beta, \quad c_4 = 1,$$

with distinct stage nodes. The generic formulas exclude degenerate parameter choices; one convenient nondegeneracy condition is

$$\alpha\beta(\alpha - 1)(\beta - 1)(\alpha - \beta)(2\alpha - 1)(6\alpha\beta - 4\alpha - 4\beta + 3) \neq 0.$$

Example 12 (The three-eighths rule). *Choosing $\alpha = 1/3$ and $\beta = 2/3$ produces the fourth-order three-eighths rule*

$$\begin{array}{c|cccc}
0 & 0 & 0 & 0 & 0 \\
\frac{1}{3} & \frac{1}{3} & 0 & 0 & 0 \\
\frac{2}{3} & -\frac{1}{3} & 1 & 0 & 0 \\
1 & 1 & -1 & 1 & 0 \\
\hline
& \frac{1}{8} & \frac{3}{8} & \frac{3}{8} & \frac{1}{8}
\end{array} .$$

Its stages are

$$\begin{aligned}
k_1 &= f(t_n, x_n), \\
k_2 &= f\left(t_n + \frac{\tau_n}{3}, x_n + \frac{\tau_n}{3}k_1\right), \\
k_3 &= f\left(t_n + \frac{2\tau_n}{3}, x_n + \tau_n\left(-\frac{1}{3}k_1 + k_2\right)\right), \\
k_4 &= f\left(t_n + \tau_n, x_n + \tau_n(k_1 - k_2 + k_3)\right), \\
x_{n+1} &= x_n + \frac{\tau_n}{8}(k_1 + 3k_2 + 3k_3 + k_4).
\end{aligned}$$

Repeated nodes give a separate, *confluent* family. Setting $c_2 = c_3 = 1/2$ and letting $\delta = a_{32} \neq 0$ yields

$$(36) \quad \begin{array}{c|cccc} 0 & 0 & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} - \delta & \delta & 0 & 0 \\ 1 & 0 & 1 - \frac{1}{2\delta} & \frac{1}{2\delta} & 0 \\ \hline & \frac{1}{6} & \frac{4\delta - 1}{6\delta} & \frac{1}{6\delta} & \frac{1}{6} \end{array}.$$

Example 13 (Classical RK4). *The choice $\delta = 1/2$ in (36) gives the classical tableau*

$$\begin{array}{c|cccc} 0 & 0 & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ \hline & \frac{1}{6} & \frac{1}{3} & \frac{1}{3} & \frac{1}{6} \end{array}.$$

Equivalently,

$$\begin{aligned} k_1 &= f(t_n, x_n), \\ k_2 &= f\left(t_n + \frac{\tau_n}{2}, x_n + \frac{\tau_n}{2}k_1\right), \\ k_3 &= f\left(t_n + \frac{\tau_n}{2}, x_n + \frac{\tau_n}{2}k_2\right), \\ k_4 &= f(t_n + \tau_n, x_n + \tau_n k_3), \\ x_{n+1} &= x_n + \frac{\tau_n}{6}(k_1 + 2k_2 + 2k_3 + k_4). \end{aligned}$$

6.4. Embedded pairs. Adaptive solvers need local error estimates. They obtain them by combining the same stages in two different ways. Given fixed A and c , choose two weight vectors b and $\widehat{b}$ and form

$$\begin{aligned} x_{n+1}^{[p]} &= x_n + \tau_n \sum_{i=1}^s b_i k_i, \\ x_{n+1}^{[p+1]} &= x_n + \tau_n \sum_{i=1}^s \widehat{b}_i k_i. \end{aligned}$$

The two formulas share all evaluations k_i , so

$$\text{err}_n = \|x_{n+1}^{[p+1]} - x_{n+1}^{[p]}\|$$

is an inexpensive local error estimate. If this quantity is well below the requested tolerance, the next step size can be increased. If it is too large, the step size is decreased and the step may be rejected.

MATLAB's `ode45` uses a seven-stage Dormand–Prince embedded pair of orders four and five. The minimal number of stages for an explicit Runge–Kutta method grows with its order; the low-order values discussed in class are

p	1	2	3	4	5	6	7	8
$s_{\min}$	1	2	3	4	6	7	9	11

6.5. The stability function. Apply a Runge–Kutta method to the linear test equation

$$(37) \quad y'(t) = \lambda y(t).$$

With $z = \tau_n \lambda$, one numerical step has the form

$$(38) \quad y_{n+1} = R(z)y_n,$$

where R is the *stability function*. The exact amplification factor is e^z .

For a Runge–Kutta method, the stage equations applied to (37) give

$$(I - zA)k = \lambda y_n \mathbf{1}.$$

Substituting the resulting stages into the update gives

$$(39) \quad R(z) = 1 + zb^T(I - zA)^{-1}\mathbf{1}.$$

If the method has order p , then necessarily

$$(40) \quad R(z) - e^z = O(z^{p+1}) \quad \text{as } z \rightarrow 0.$$

For example, the eight fourth-order conditions imply

$$R(z) = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \frac{z^4}{4!} + O(z^5).$$

Condition (40) is necessary for order p , but it only tests a linear ODE and is not sufficient to establish order p for general nonlinear equations.

7. SEPTEMBER 9, 2026

This lecture introduces extrapolation as a way to increase the order of an existing time-stepping method. We then begin the study of stability as a property of solutions of the differential equation itself.

7.1. Accuracy and computational cost. Consider a method of order p on the uniform grid

$$t_n = t_0 + n\tau, \quad n = 0, \dots, N, \quad t_N = T,$$

and let $Y_\tau(T) = x_N$ denote the numerical approximation at the final time. Convergence of order p means that, for sufficiently small τ ,

$$(41) \quad \|Y_\tau(T) - x(T)\| \leq C(T)\tau^p.$$

In particular, $Y_\tau(T) \rightarrow x(T)$ as $\tau \rightarrow 0$. For a fixed step size, increasing the order generally decreases the error, provided that τ is in the asymptotic regime in which (41) is valid.

Suppose a tolerance $\varepsilon > 0$ is prescribed. Balancing the error estimate with the tolerance gives

$$C(T)\tau^p \approx \varepsilon, \quad \tau \approx \left(\frac{\varepsilon}{C(T)} \right)^{1/p}.$$

Since $N = (T - t_0)/\tau$, the resulting work, measured by the number of time steps, is approximately

$$(42) \quad N \approx (T - t_0) \left(\frac{C(T)}{\varepsilon} \right)^{1/p}.$$

Thus a higher-order method has a more favorable dependence on the tolerance. Formula (42) does not include the work per step or the size of the error constant, both of which also matter in practice.

7.2. Richardson extrapolation. Extrapolation requires more than an error bound. Assume the stronger asymptotic expansion

$$(43) \quad Y_\tau(T) = x(T) + c_p(T)\tau^p + c_{p+1}(T)\tau^{p+1} + c_{p+2}(T)\tau^{p+2} + \dots.$$

Rerunning the same method with half the step size gives

$$Y_{\tau/2}(T) = x(T) + c_p(T)\frac{\tau^p}{2^p} + c_{p+1}(T)\frac{\tau^{p+1}}{2^{p+1}} + O(\tau^{p+2}).$$

Multiplying this equation by 2^p and subtracting (43) cancels the leading error term:

$$2^p Y_{\tau/2}(T) - Y_\tau(T) = (2^p - 1)x(T) + O(\tau^{p+1}).$$

Consequently, the *Richardson extrapolate*

$$(44) \quad Y_\tau^{[1]}(T) = \frac{2^p Y_{\tau/2}(T) - Y_\tau(T)}{2^p - 1}$$

satisfies

$$Y_\tau^{[1]}(T) = x(T) + O(\tau^{p+1}).$$

This gains one order. The underlying one-step method is unchanged; the additional accuracy comes from combining two complete runs.

7.2.1. Symmetric methods. For an autonomous problem, a one-step method is *symmetric* if a step of length $-\tau$ exactly reverses a step of length τ :

$$\Psi^{-\tau} = (\Psi^\tau)^{-1}.$$

Symmetric methods have even order. Under the usual smoothness assumptions, their fixed-time global error expansion contains powers of one parity. If the method has order p , the expansion takes the form

$$(45) \quad Y_\tau(T) = x(T) + d_p(T)\tau^p + d_{p+2}(T)\tau^{p+2} + d_{p+4}(T)\tau^{p+4} + \dots.$$

Applying the same combination as in (44) cancels the τ^p term, but now there is no τ^{p+1} term. Hence

$$\frac{2^p Y_{\tau/2}(T) - Y_\tau(T)}{2^p - 1} = x(T) + O(\tau^{p+2}),$$

so one extrapolation raises the order by two.

7.3. Extrapolation from several step sizes. The preceding cancellation can be repeated using several numerical solutions. Choose distinct positive integers $n_1, \dots, n_m$, set

$$\tau_i = \frac{\bar{\tau}}{n_i}, \quad i = 1, \dots, m,$$

and compute $Y_{\tau_i}(T)$ for every i . Motivated by (43), fit the generalized polynomial

$$(46) \quad P(h) = a_0 + \sum_{j=0}^{m-2} a_{p+j} h^{p+j}$$

to these values:

$$(47) \quad P(\tau_i) = Y_{\tau_i}(T), \quad i = 1, \dots, m.$$

The unknowns $a_0, a_p, \dots, a_{p+m-2}$ are determined by the linear system

$$\begin{bmatrix} 1 & \tau_1^p & \tau_1^{p+1} & \dots & \tau_1^{p+m-2} \\ 1 & \tau_2^p & \tau_2^{p+1} & \dots & \tau_2^{p+m-2} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & \tau_m^p & \tau_m^{p+1} & \dots & \tau_m^{p+m-2} \end{bmatrix} \begin{bmatrix} a_0 \\ a_p \\ a_{p+1} \\ \vdots \\ a_{p+m-2} \end{bmatrix} = \begin{bmatrix} Y_{\tau_1}(T) \\ Y_{\tau_2}(T) \\ \vdots \\ Y_{\tau_m}(T) \end{bmatrix}.$$

For a vector-valued ODE, the same coefficient matrix is used for every component. Extrapolating to zero step size gives

$$(48) \quad \bar{x}(T) = P(0) = a_0.$$

The first $m - 1$ error terms have been removed, so, under the assumed full expansion,

$$\bar{x}(T) = x(T) + O(\bar{\tau}^{p+m-1}).$$

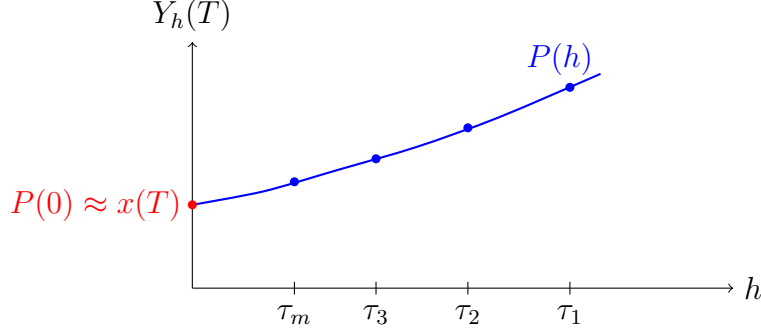
Equivalently, a_0 is a linear combination of the computed values,

$$a_0 = \sum_{i=1}^m \gamma_i Y_{\tau_i}(T),$$

where the weights are chosen so that

$$\sum_{i=1}^m \gamma_i = 1, \quad \sum_{i=1}^m \gamma_i \tau_i^{p+j} = 0, \quad j = 0, \dots, m-2.$$

For $m = 2$, $n_1 = 1$, and $n_2 = 2$, these conditions recover (44). For a symmetric error expansion, one instead uses the powers $p, p+2, p+4, \dots$ in (46).



7.4. Stability of solutions. Return to the IVP

$$(49) \quad x'(t) = f(t, x(t)), \quad x(t_0) = x_0,$$

and write its solution as $x(t; t_0, x_0)$ to display its dependence on the initial data.

Definition 4 (Stability of a solution). *The solution through x_0 is stable if, for every $\varepsilon > 0$, there exists $\delta > 0$ such that*

$$\|x_0 - y_0\| < \delta \implies \|x(t; t_0, x_0) - x(t; t_0, y_0)\| < \varepsilon$$

for every $t \geq t_0$.

The estimate from Grönwall's inequality in (7) gives continuous dependence over every fixed finite interval. Stability is stronger: the same δ must control the separation for all future times.

Definition 5 (Asymptotic stability). *A stable solution through x_0 is asymptotically stable if all sufficiently close initial values y_0 also satisfy*

$$\|x(t; t_0, x_0) - x(t; t_0, y_0)\| \longrightarrow 0 \quad \text{as } t \rightarrow \infty.$$

For an autonomous equation $x' = f(x)$, a point x_* satisfying

$$f(x_*) = 0$$

is an *equilibrium*. The solution starting at the equilibrium is constant: if $x_0 = x_*$, then $x(t; t_0, x_*) = x_*$ for all $t \geq t_0$. Stability of this equilibrium is obtained from the preceding definitions by comparing this constant solution with solutions starting at nearby points.

Example 14 (Linear autonomous system). *Consider*

$$x' = Ax.$$

The origin is an equilibrium, and

$$x(t; t_0, x_0) = e^{(t-t_0)A}x_0.$$

Thus the difference of two solutions is

$$x(t; t_0, x_0) - x(t; t_0, y_0) = e^{(t-t_0)A}(x_0 - y_0).$$

The system is stable precisely when e^{tA} remains bounded for $t \geq 0$, and it is asymptotically stable precisely when $e^{tA} \rightarrow 0$. In spectral terms, stability requires every eigenvalue to have nonpositive real part and every eigenvalue on the imaginary axis to be semisimple. Asymptotic stability holds exactly when every eigenvalue has strictly negative real part.

In the scalar case $x' = \lambda x$, the solution is $x(t) = e^{\lambda(t-t_0)}x_0$. Hence the equilibrium at the origin is stable when $\operatorname{Re} \lambda \leq 0$, and it is asymptotically stable when $\operatorname{Re} \lambda < 0$.

These notions describe the behavior of the exact differential equation. They are distinct from numerical absolute stability, which asks whether a time-stepping method reproduces the decay of a stable test equation.

8. SEPTEMBER 11, 2026

We continue the discussion of stability from the previous lecture. The main goals are to linearize an autonomous ODE near an equilibrium, to study the resulting system using the matrix exponential, and to extend the solution formula to time-dependent linear systems by means of the Dyson series.

8.1. Linearization near an equilibrium. Consider the autonomous nonlinear equation

$$(50) \quad x' = f(x)$$

and let x_* be an equilibrium, so that $f(x_*) = 0$. Introduce the perturbation

$$y = x - x_*.$$

Since x_* is constant, $y' = x'$. If f is continuously differentiable, its expansion near x_* is

$$f(x_* + y) = f(x_*) + f_x(x_*)y + r(y) = Ay + r(y), \quad A = f_x(x_*),$$

where

$$\frac{\|r(y)\|}{\|y\|} \rightarrow 0 \quad \text{as } y \rightarrow 0.$$

Thus the exact perturbation equation is

$$(51) \quad y' = Ay + r(y),$$

and its linearization is

$$(52) \quad y' = Ay.$$

The equilibrium x_* corresponds to the zero solution of the perturbation equation. Hence studying whether $x(t)$ remains close to x_* begins with studying whether solutions of (52) remain close to zero.

8.2. Autonomous linear systems and the matrix exponential.

Consider the initial-value problem

$$(53) \quad y' = Ay, \quad y(t_0) = y_0,$$

where $A \in \mathbb{R}^{n \times n}$ is constant. Repeated differentiation gives

$$y^{(k)}(t_0) = A^k y_0.$$

The Taylor series about t_0 therefore gives

$$y(t) = y_0 + (t - t_0)Ay_0 + \frac{(t - t_0)^2}{2!}A^2y_0 + \cdots$$

$$= \left(\sum_{k=0}^{\infty} \frac{(t-t_0)^k A^k}{k!} \right) y_0.$$

Definition 6 (Matrix exponential). *For a square matrix B , its exponential is*

$$(54) \quad e^B = \sum_{k=0}^{\infty} \frac{B^k}{k!}.$$

The series converges in every matrix norm. With this notation, the solution of (53) is

$$(55) \quad y(t) = e^{(t-t_0)A} y_0.$$

Indeed,

$$\frac{d}{dt} e^{(t-t_0)A} = A e^{(t-t_0)A} = e^{(t-t_0)A} A, \quad e^{0A} = I.$$

As observed in the previous lecture, the zero solution of (53) is stable exactly when

$$(56) \quad \sup_{t \geq 0} \|e^{tA}\| < \infty,$$

and it is asymptotically stable exactly when

$$(57) \quad \|e^{tA}\| \longrightarrow 0 \quad \text{as } t \rightarrow \infty.$$

These statements follow directly from

$$y(t; t_0, y_0) - y(t; t_0, z_0) = e^{(t-t_0)A} (y_0 - z_0).$$

Linearity also implies that stability of the zero solution is equivalent to stability of every solution.

8.3. Time-dependent linear systems and the Dyson series. Now consider the nonautonomous linear equation

$$(58) \quad y' = A(t)y, \quad y(t_0) = y_0,$$

where $A(t)$ is continuous. Its integral equation is

$$(59) \quad y(t) = y_0 + \int_{t_0}^t A(t_1)y(t_1) dt_1.$$

At an intermediate time t_1 , the same equation reads

$$y(t_1) = y_0 + \int_{t_0}^{t_1} A(t_2)y(t_2) dt_2.$$

Substituting this formula into (59) gives

$$\begin{aligned} y(t) &= y_0 + \left(\int_{t_0}^t A(t_1) dt_1 \right) y_0 \\ &\quad + \int_{t_0}^t \int_{t_0}^{t_1} A(t_1)A(t_2)y(t_2) dt_2 dt_1. \end{aligned}$$

Continuing this substitution produces the *Dyson series*

$$(60) \quad y(t) = U(t, t_0)y_0,$$

where the evolution operator is

$$(61) \quad U(t, t_0) = I + \sum_{k=1}^{\infty} \int_{t_0}^t \int_{t_0}^{t_1} \cdots \int_{t_0}^{t_{k-1}} A(t_1)A(t_2) \cdots A(t_k) dt_k \cdots dt_2 dt_1.$$

If $\|A(s)\| \leq M$ on $[t_0, t]$, the norm of the k -th term is at most

$$\frac{M^k(t - t_0)^k}{k!},$$

which proves convergence of the series. Termwise differentiation shows that

$$\frac{\partial}{\partial t} U(t, t_0) = A(t)U(t, t_0), \quad U(t_0, t_0) = I.$$

Because matrices evaluated at different times need not commute, the order $t_1 \geq t_2 \geq \cdots \geq t_k$ in (61) is essential. A common shorthand is the *time-ordered exponential*

$$(62) \quad U(t, t_0) = \mathcal{T} \exp \left(\int_{t_0}^t A(s) ds \right).$$

If $A(s)A(r) = A(r)A(s)$ for every r, s , time ordering is unnecessary and

$$U(t, t_0) = \exp \left(\int_{t_0}^t A(s) ds \right).$$

In particular, when $A(t) \equiv A$, the ordered integration region has volume $(t - t_0)^k/k!$, and the Dyson series reduces to $e^{(t-t_0)A}$.

8.4. Properties of the matrix exponential. The power-series definition immediately gives several useful identities.

Proposition 1. *Let A and B be square matrices of the same size.*

(i) *If M is invertible, then*

$$e^{MAM^{-1}} = Me^AM^{-1}.$$

(ii) *If $AB = BA$, then*

$$e^{A+B} = e^Ae^B = e^Be^A.$$

(iii) *For a block-diagonal matrix,*

$$e^{\text{diag}(A_1, \dots, A_m)} = \text{diag}(e^{A_1}, \dots, e^{A_m}).$$

Proof. For the first identity, observe that $(MAM^{-1})^k = MA^kM^{-1}$ and apply (54). The third identity follows similarly because powers preserve block-diagonal structure. When A and B commute, the binomial theorem is valid for $(A+B)^k$. Substitution into the exponential series and collection of like terms gives the second identity. $\square$

8.5. Jordan form and linear stability. The preceding identities reduce the stability analysis of $y' = Ay$ to the analysis of individual Jordan blocks. Over $\mathbb{C}$, write

$$(63) \quad A = PJP^{-1}, \quad J = \text{diag}(J_1, \dots, J_m).$$

If J_k is a Jordan block of size q_k associated with the eigenvalue λ_k , then

$$J_k = \lambda_k I + N_k, \quad N_k^{q_k} = 0.$$

Since $\lambda_k I$ commutes with N_k ,

$$(64) \quad e^{tJ_k} = e^{t\lambda_k} \left(I + tN_k + \frac{t^2 N_k^2}{2!} + \dots + \frac{t^{q_k-1} N_k^{q_k-1}}{(q_k-1)!} \right).$$

Thus each Jordan block contributes an exponential factor multiplied by a polynomial in t .

Theorem 1 (Stability criterion for a linear system). *For $y' = Ay$, the following statements hold.*

(i) *The zero solution is asymptotically stable if and only if*

$$\operatorname{Re} \lambda < 0 \quad \text{for every } \lambda \in \sigma(A).$$

(ii) *The zero solution is stable if and only if every eigenvalue satisfies $\operatorname{Re} \lambda \leq 0$ and every eigenvalue on the imaginary axis is semisimple, meaning that all of its Jordan blocks have size one.*

Proof. By Proposition 1 and (63),

$$e^{tA} = P e^{tJ} P^{-1} = P \operatorname{diag}(e^{tJ_1}, \dots, e^{tJ_m}) P^{-1}.$$

If $\operatorname{Re} \lambda_k < 0$, the exponential decay in (64) dominates every polynomial factor, so the block tends to zero. If $\operatorname{Re} \lambda_k > 0$, the block has an exponentially growing solution. When $\operatorname{Re} \lambda_k = 0$, the scalar exponential has constant magnitude. The block is therefore bounded precisely when its nilpotent part is zero, or equivalently when its size is one. Combining the blocks with (56) and (57) proves both assertions. $\square$

Example 15 (A defective eigenvalue on the imaginary axis). *Let*

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

The only eigenvalue is zero, but its Jordan block has size two. Since $A^2 = 0$,

$$e^{tA} = I + tA = \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}.$$

Hence

$$y_2(t) = y_2(0), \quad y_1(t) = y_1(0) + t y_2(0).$$

Arbitrarily small initial data with $y_2(0) \neq 0$ lead to linear growth, so the origin is not stable. This illustrates why nonpositive real parts alone are insufficient for stability.

For the nonlinear perturbation equation (51), the linearization principle gives two decisive cases. If the spectrum of $f_x(x_*)$ lies in the open left half-plane, then x_* is locally asymptotically stable. If the spectrum contains an eigenvalue with positive real part, then x_* is unstable. If it meets the imaginary axis but not the open right half-plane, linearization alone may be inconclusive; the nonlinear remainder $r(y)$ can determine the behavior.

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We develop two complementary approaches to stability. Linearization transfers spectral information from a Jacobian matrix to a nonlinear equilibrium, while Lyapunov functions establish stability without computing the solution explicitly. We then apply the same ideas to time-dependent linear equations and to the discrete evolution produced by Runge–Kutta methods.

9.1. From linear to nonlinear stability. Recall the autonomous equation

$$x' = f(x),$$

an equilibrium x_* satisfying $f(x_*) = 0$, and its Jacobian

$$A = f_x(x_*).$$

With $y = x - x_*$, the perturbation equation derived in (51) is

$$y' = Ay + r(y), \quad \frac{\|r(y)\|}{\|y\|} \longrightarrow 0 \quad \text{as } y \rightarrow 0.$$

The matrix A is called *Hurwitz* if all of its eigenvalues lie in the open left half-plane:

$$\operatorname{Re} \lambda < 0 \quad \text{for every } \lambda \in \sigma(A).$$

The linearization theorem gives the following nonlinear conclusion.

Theorem 2 (Linearization criterion). *Suppose f is continuously differentiable near x_* .*

- (i) *If $f_x(x_*)$ is Hurwitz, then x_* is locally asymptotically stable.*
- (ii) *If $f_x(x_*)$ has an eigenvalue with positive real part, then x_* is unstable.*

If no eigenvalue of $f_x(x_*)$ lies on the imaginary axis, the equilibrium is *hyperbolic*. The Hartman–Grobman theorem says that the nonlinear flow near a hyperbolic equilibrium is locally topologically equivalent to its linearization $y' = Ay$. If the spectrum meets the imaginary axis, the linearized system alone may not determine stability.

9.2. Lyapunov's direct method. Eigenvalue calculations or explicit solution formulas can be expensive. Lyapunov's method instead measures the size of a solution by a scalar energy-like function.

Definition 7 (Lyapunov function). *A continuously differentiable function V , defined near an equilibrium x_* , is a Lyapunov function if*

$$V(x_*) = 0, \quad V(x) > 0 \quad \text{for } x \neq x_*$$

near x_ , and its derivative along solutions,*

$$(65) \quad \dot{V}(x) = \nabla V(x)^\top f(x),$$

is nonpositive.

Theorem 3 (Lyapunov stability theorem). *If V is positive definite near x_* and $\dot{V} \leq 0$, then x_* is stable. If, in addition, $\dot{V}(x) < 0$ for every $x \neq x_*$ sufficiently close to x_* , then x_* is locally asymptotically stable.*

The idea is that a solution cannot cross outward through a sufficiently small level set of V when $V(x(t))$ is nonincreasing. Strict decrease forces the solution toward the equilibrium.

9.3. Quadratic Lyapunov functions for linear systems. Consider again

$$(66) \quad y' = Ay.$$

Let Q be a real symmetric positive-definite matrix and define

$$(67) \quad V(y) = y^\top Q y.$$

If $\lambda_{\min}(Q)$ and $\lambda_{\max}(Q)$ denote the extreme eigenvalues of Q , then

$$(68) \quad \lambda_{\min}(Q)\|y\|^2 \leq V(y) \leq \lambda_{\max}(Q)\|y\|^2.$$

Consequently, $V(y(t)) \rightarrow 0$ if and only if $y(t) \rightarrow 0$, and boundedness of $V(y(t))$ is equivalent to boundedness of $y(t)$.

Along a solution of (66),

$$\begin{aligned} \frac{d}{dt}V(y(t)) &= y'(t)^\top Q y(t) + y(t)^\top Q y'(t) \\ (69) \qquad \qquad &= y(t)^\top (A^\top Q + Q A) y(t). \end{aligned}$$

If

$$A^\top Q + Q A \preceq -\mu I \quad \text{for some } \mu > 0,$$

then

$$\begin{aligned} \dot{V}(y(t)) &\leq -\mu \|y(t)\|^2 \\ &\leq -\frac{\mu}{\lambda_{\max}(Q)} V(y(t)). \end{aligned}$$

Grönwall's inequality therefore gives

$$(70) \qquad V(y(t)) \leq e^{-\mu(t-t_0)/\lambda_{\max}(Q)} V(y(t_0)),$$

so the origin is exponentially stable.

9.3.1. *Lyapunov equation.* Choose a symmetric positive-definite matrix C . The matrix equation

$$(71) \qquad A^\top Q + Q A = -C$$

is called the *continuous Lyapunov equation*. Often one simply takes $C = I$ and solves for Q .

Theorem 4 (Lyapunov matrix theorem). *The matrix A is Hurwitz if and only if, for every symmetric $C \succ 0$, equation (71) has a unique symmetric solution $Q \succ 0$.*

Proof. Suppose first that A is Hurwitz. By the stability criterion from the previous lecture, $e^{tA} \rightarrow 0$ exponentially. Define

$$(72) \qquad Q = \int_0^\infty e^{tA^\top} C e^{tA} dt.$$

The integral converges, Q is symmetric, and for $v \neq 0$,

$$v^\top Q v = \int_0^\infty (e^{tA} v)^\top C (e^{tA} v) dt > 0.$$

Thus $Q \succ 0$. Moreover,

$$\begin{aligned} A^\top Q + QA &= \int_0^\infty \frac{d}{dt} \left(e^{tA^\top} C e^{tA} \right) dt \\ &= \left[e^{tA^\top} C e^{tA} \right]_{t=0}^{t=\infty} = -C. \end{aligned}$$

If Q_1 and Q_2 are two solutions, their difference D satisfies $A^\top D + DA = 0$. Hence $e^{tA^\top} D e^{tA} = D$ for all t , while the left-hand side tends to zero. Therefore $D = 0$, proving uniqueness.

Conversely, suppose $Q \succ 0$ and $C \succ 0$ satisfy (71). For a possibly complex eigenpair $Av = \lambda v$, multiplication by the conjugate transpose v^* gives

$$2 \operatorname{Re}(\lambda) v^* Q v = -v^* C v < 0.$$

Since $v^* Q v > 0$, every eigenvalue has negative real part. Thus A is Hurwitz. $\square$

The integral formula shows the equivalence between the spectral and Lyapunov approaches to linear stability. It also proves the nonlinear stability conclusion locally. Indeed, choose $C = I$ and use the same quadratic function for $y' = Ay + r(y)$. Then

$$\dot{V}(y) = -\|y\|^2 + 2y^\top Q r(y) = -\|y\|^2 + o(\|y\|^2),$$

which is negative for all sufficiently small nonzero y .

9.4. A time-dependent linear system. Consider

$$(73) \quad y' = A(t)y.$$

For the simple Lyapunov function

$$V(y) = \frac{1}{2} y^\top y = \frac{1}{2} \|y\|^2,$$

we have

$$(74) \quad \dot{V}(y(t)) = y(t)^\top S(t) y(t), \quad S(t) = \frac{1}{2} (A(t) + A(t)^\top).$$

Thus $S(t) \preceq 0$ for all $t \geq t_0$ is sufficient for stability, because the Euclidean norm cannot increase. The uniform condition

$$(75) \quad S(t) \preceq -\alpha I, \quad t \geq t_0, \quad \alpha > 0,$$

implies

$$\dot{V} \leq -2\alpha V, \quad \|y(t)\| \leq e^{-\alpha(t-t_0)}\|y(t_0)\|,$$

and hence uniform exponential stability. Pointwise negative definiteness without a uniform bound does not by itself guarantee decay to zero.

9.5. Runge–Kutta methods on a linear system. We now examine the numerical evolution of the constant-coefficient system $y' = Ay$ with a fixed step size τ .

9.5.1. *Forward Euler and an explicit midpoint method.* Forward Euler has one stage,

$$k_1 = Ay_n,$$

and therefore

$$(76) \quad y_{n+1} = y_n + \tau k_1 = (I + \tau A)y_n.$$

The two-stage explicit midpoint method is

$$\begin{aligned} k_1 &= Ay_n, \\ k_2 &= A\left(y_n + \frac{\tau}{2}k_1\right), \\ y_{n+1} &= y_n + \tau k_2. \end{aligned}$$

Consequently,

$$(77) \quad y_{n+1} = \left(I + \tau A + \frac{\tau^2}{2}A^2\right)y_n.$$

Every explicit two-stage Runge–Kutta method of order two has the same update matrix on this linear equation, even though its internal stages may differ.

More generally, recall the Runge–Kutta stability function R from (39). Applied to $y' = Ay$, a Runge–Kutta method gives

$$(78) \quad y_{n+1} = G(\tau A)y_n, \quad G(\tau A) = R(\tau A).$$

For an explicit s -stage method, R is a polynomial of degree at most s . If that method also has order s , then

$$R(z) = \sum_{j=0}^s \frac{z^j}{j!},$$

so $G(\tau A)$ is the degree- s Taylor approximation of $e^{\tau A}$.

9.6. Stability of a discrete linear evolution. Consider a general recurrence

$$(79) \quad y_{n+1} = G y_n, \quad y_n = G^n y_0.$$

The zero solution is stable precisely when $\{G^n\}_{n \geq 0}$ is bounded, and it is asymptotically stable precisely when $G^n \rightarrow 0$.

As in the continuous case, write $G = PJP^{-1}$ over $\mathbb{C}$. For a Jordan block

$$J_k = \mu_k I + N_k, \quad N_k^{q_k} = 0,$$

the binomial theorem gives

$$(80) \quad J_k^n = \sum_{j=0}^{\min(n, q_k-1)} \binom{n}{j} \mu_k^{n-j} N_k^j.$$

The binomial coefficient grows polynomially in n . It is dominated by $|\mu_k|^n$ when $|\mu_k| < 1$, but produces growth on the unit circle when $N_k \neq 0$.

Theorem 5 (Stability criterion for a discrete linear system). *For the recurrence (79):*

(i) *the zero solution is asymptotically stable if and only if*

$$\rho(G) = \max_{\mu \in \sigma(G)} |\mu| < 1;$$

(ii) *the zero solution is stable if and only if $\rho(G) \leq 1$ and every eigenvalue with $|\mu| = 1$ is semisimple.*

For a Runge–Kutta method, the spectral mapping theorem applied to (78) gives

$$\sigma(R(\tau A)) = \{R(\tau \lambda) : \lambda \in \sigma(A)\}.$$

Therefore the numerical solution is asymptotically stable when

$$(81) \quad |R(\tau\lambda)| < 1 \quad \text{for every } \lambda \in \sigma(A).$$

This motivates the *absolute stability region*

$$(82) \quad \mathcal{S} = \{z \in \mathbb{C} : |R(z)| < 1\}.$$

For example, forward Euler has $R(z) = 1 + z$. If $\lambda < 0$ is real, then $|1 + \tau\lambda| < 1$ is equivalent to

$$0 < \tau < \frac{2}{|\lambda|}.$$

Thus even when the differential equation is asymptotically stable, the step size must place every scaled eigenvalue $\tau\lambda$ inside the method's absolute stability region.